XMODEM Gateway

User Manual

Version 0.3.3

Ricky Bryce

A telnet-based XMODEM file transfer server, SSH gateway, Hayes-compatible modem emulator, text-mode web browser, and AI chat client for retro hardware — PETSCII, ANSI, and ASCII terminals.

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1. Introduction

XMODEM Gateway is a standalone server written in Rust that bridges retro hardware and modern networks. It accepts telnet and SSH connections, speaks the XMODEM / XMODEM-1K / YMODEM file-transfer protocols, emulates a Hayes AT modem on a physical serial port, proxies outbound connections to remote telnet and SSH servers, renders web pages as plain text, looks up the weather, and provides a terminal AI-chat front end for the Groq API.

It was designed to let 40-year-old computers use 40-minute-old networks. The telnet interface is the headline feature because it works out of the box with any machine that can speak a serial stream: Commodore 64, CP/M systems, the AltairDuino, RC2014 / SC126, and countless other homebrew boards. Once the server is running on a PC, connect from anywhere on your LAN:

```
telnet 192.168.1.160 2323
```

Or from a serial-attached retro machine, dial through the emulator:

```
ATDT 192.168.1.160:2323
```

1.1 Supported Terminal Types

On every inbound connection the gateway asks the user to press `Backspace` so it can auto-detect the terminal. The byte received uniquely identifies the client:

Byte	Terminal	Typical hardware
`0x14`	PETSCII	Commodore 64, VIC-20, C128 in 40-column mode
`0x08` or `0x7F`	ANSI	Modern terminal emulators, PuTTY, xterm, macOS Terminal
anything else	ASCII	Dumb terminals, line printers, bridge boards without colour

After terminal detection the user is asked whether to enable colour. There is no default — the user must explicitly press `Y` or `N`. Colour output uses PETSCII colour codes on the Commodore, ANSI escape sequences on modern terminals, and plain text on ASCII.

1.2 Feature Overview

- **File transfer** — XMODEM (128-byte), XMODEM-1K, and YMODEM with auto-detection on upload and a selection prompt on download.
- **Hayes AT modem emulator** — real serial port, full command set, S-registers S0–S26, stored phone-number slots, and `+++` escape.
- **SSH server** — optional encrypted access with its own credentials and a persistent Ed25519 host key.

- **SSH gateway** — outbound proxy for SSH servers, with TOFU host-key trust and Ed25519 public-key auth.
- **Telnet gateway** — outbound proxy with RFC-1143 Q-method option negotiation or a raw-TCP escape hatch.
- **Web browser** — text-mode HTTP/HTTPS/Gopher with forms, bookmarks, and 40-column-friendly layout.
- **AI chat** — Groq API, `llama-3.3-70b-versatile`, paginated answers wrapped to the terminal width.
- **Weather** — 3-day forecast from Open-Meteo for any US ZIP code.
- **GUI console** — eframe/egui configuration editor with live server log.

> **Security notice**
>
> Telnet transmits all data, including passwords, in cleartext. The telnet interface is intended for *local or private network use only*. Use the SSH interface for anything leaving your LAN, and keep `xmodem.conf` at mode `0600` because it contains plaintext credentials.

2. Installation

2.1 Linux AppImage (pre-built)

The simplest installation on a modern Linux distribution is the pre-built AppImage that ships with every release:

```
chmod +x XMODEM_Gateway-x86_64.AppImage
./XMODEM_Gateway-x86_64.AppImage
```

AppImages bundle their own libraries and run on any mainstream glibc-based distribution without installation. On first run the AppImage creates `xmodem.conf` in the current working directory.

> **Verification**
> Each release publishes a SHA-256 checksum, an optional GPG signature, and a Sigstore keyless signature bound to the publisher's GitHub identity. Verify at least the SHA-256 before running any binary from the internet: `sha256sum -c xmodem-gateway-v0.3.3-…tar.gz.sha256`.

2.2 Building from Source

The project is written in pure Rust using the 2024 edition. The minimum supported compiler is Rust **1.87**.

2.2.1 Debian 13 / Ubuntu

```
sudo apt update
sudo apt install -y build-essential pkg-config cmake curl libudev-dev
curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh
source "$HOME/.cargo/env"
```

2.2.2 Fedora / RHEL / AlmaLinux

```
sudo dnf install -y gcc gcc-c++ make cmake pkg-config curl
curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh
source "$HOME/.cargo/env"
```

2.2.3 Arch Linux

```
sudo pacman -S --needed base-devel cmake curl
curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh
source "$HOME/.cargo/env"
```

2.2.4 macOS

```
xcode-select --install
curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh
source "$HOME/.cargo/env"
brew install cmake
```

2.2.5 Windows

1. Run the rustup installer from `https://rustup.rs`.
2. When prompted, install the Visual Studio C++ Build Tools.
3. Install CMake from `https://cmake.org/download/` or via `winget install Kitware.CMake`.
4. Open a fresh PowerShell or cmd.exe so rustup is on your `PATH`.

2.2.6 Verify the toolchain

```
rustc --version    # should show 1.87 or later
cargo --version
cmake --version
```

2.2.7 Compile

```
cargo build --release
```

The resulting binary is at `target/release/xmodem-gateway`. The first full build pulls a couple of hundred crates and may take 3–8 minutes depending on your machine; subsequent incremental builds are quick.

2.2.8 Key dependencies

The gateway pulls the following well-known crates as direct dependencies: `tokio` (async runtime), `russh` (SSH server and client), `serialport` (cross-platform serial I/O), `ureq` (blocking HTTP client used by the browser and AI chat), `html2text` (HTML to plain-text rendering), `eframe` / `egui_extras` (GUI), `signal-hook` (POSIX signals), and `serde_json`. Full transitive list is in `Cargo.lock`.

2.3 Running as a systemd Service (Linux)

A hardened unit file is bundled at `contrib/systemd/xmodem-gateway.service`. To install:

```
# Create a dedicated unprivileged service user
sudo useradd --system --home-dir /var/lib/xmodem-gateway \
             --shell /usr/sbin/nologin xmodem-gateway
sudo install -d -m 0750 -o xmodem-gateway -g xmodem-gateway \
             /var/lib/xmodem-gateway

# Install the binary
sudo install -m 0755 target/release/xmodem-gateway /usr/local/bin/

# Install and enable the unit
sudo install -m 0644 contrib/systemd/xmodem-gateway.service \
             /etc/systemd/system/
sudo systemctl daemon-reload
sudo systemctl enable --now xmodem-gateway.service

# Watch the log
journalctl -u xmodem-gateway -f
```

The unit ships with defensive hardening: `NoNewPrivileges`, `PrivateTmp`, `ProtectSystem=strict`, `ProtectHome`, namespace restrictions, `SystemCallFilter=@system-service`, an empty capability bounding set, a 512 MiB memory cap, and `TasksMax=256`. The service user has no shell, no home directory, and write access only to `/var/lib/xmodem-gateway`.

If you want to listen on privileged ports (telnet on 23 or SSH on 22), uncomment the `CapabilityBoundingSet=CAP_NET_BIND_SERVICE` and matching `AmbientCapabilities` lines in the unit.

`systemctl reload xmodem-gateway` sends SIGHUP, which the binary handles as a graceful stop. Restart the service to pick up config changes.

3. First Run & Configuration File

On the very first launch the gateway looks for `xmodem.conf` in the current working directory. If the file does not exist it is created with sensible defaults and a short log message is printed. On subsequent launches the file is read, every missing key is filled in with its default, and the file is rewritten so that future editors see the full set of keys and comments.

> **Unix permissions**
>
> `xmodem.conf` holds plaintext credentials (telnet password, SSH password, Groq API key). On Unix the file is set to mode `0600` (owner-only read/write) on every write. The same applies to `dialup.conf`, `gateway_hosts`, `xmodem_ssh_host_key`, and `xmodem_gateway_ssh_key`.

3.1 File Format

The file is a simple `key = value` text file. Blank lines and lines starting with `#` are comments and ignored. Values are trimmed of surrounding whitespace. Unknown keys are silently ignored, so you can annotate your own notes as needed. Boolean values accept `true` / `false` in any case.

```
# XMODEM Gateway Configuration
telnet_enabled = true
telnet_port = 2323
ssh_enabled  = false
security_enabled = false
```

3.2 Configuration Key Overview

The table below summarises every key recognised by the parser. Full details including sanitisation, range checks, and live-edit behaviour are in Appendix A. The reserved-zero S-registers and stored phone-number slots are covered in chapter 9.

Key	Type	Default	Description
`telnet_enabled`	bool	`true`	Run the telnet server. Set `false` for SSH-only mode.
`telnet_port`	u16	`2323`	Telnet listen port. Must be ≥ 1.
`telnet_gateway_negotiate`	bool	`false`	Outbound gateway sends proactive `WILL TTYPE` / `WILL NAWS` / `DO ECHO`.
`telnet_gateway_raw`	bool	`false`	Outbound gateway bypasses all IAC parsing. Supersedes negotiate.
`enable_console`	bool	`true`	Open the GUI at startup. Set `false` for headless systemd operation.
`security_enabled`	bool	`false`	Require username/password login on telnet and SSH.
`username`, `password`	string	`admin`, `changeme`	Telnet credentials.
`ssh_username`, `ssh_password`	string	`admin`, `changeme`	SSH credentials, independent of telnet.
`ssh_enabled`	bool	`false`	Run the SSH server on `ssh_port`.
`ssh_port`	u16	`2222`	SSH listen port.
`transfer_dir`	string	`transfer`	Base directory for file transfers, relative to the working dir.
`max_sessions`	usize	`50`	Concurrent telnet session cap.
`idle_timeout_secs`	u64	`900`	Session idle timeout. 0 disables.
`groq_api_key`	string	empty	Key for AI chat. Empty disables the feature.
`browser_homepage`	string	`http://telnetbible.com`	Page loaded when the browser opens.
`weather_zip`	string	empty	Last-used US ZIP. Updated automatically.
`verbose`	bool	`false`	Detailed XMODEM protocol diagnostics.
`xmodem_negotiation_timeout`	u64	`45`	Seconds to wait for the far-end sender/receiver to start.
`xmodem_block_timeout`	u64	`20`	Per-block timeout.
`xmodem_max_retries`	usize	`10`	NAK retry cap per block.
`serial_enabled`	bool	`false`	Run the Hayes modem emulator on the configured serial port.

Key	Type	Default	Description
`serial_port`	string	empty	Device path (e.g. `/dev/ttyUSB0`, `COM3`).
`serial_baud`	u32	`9600`	Baud rate. Must be ≥ 300.
`serial_databits`	u8	`8`	Data bits: 5, 6, 7, or 8.
`serial_parity`	string	`none`	`none` / `odd` / `even`.
`serial_stopbits`	u8	`1`	1 or 2.
`serial_flowcontrol`	string	`none`	`none` / `hardware` / `software`.
`serial_echo`	bool	`true`	Saved `ATE` state. Restored by `ATZ`.
`serial_verbose`	bool	`true`	Saved `ATV` state.
`serial_quiet`	bool	`false`	Saved `ATQ` state.
`serial_s_regs`	string	see ch. 9	Comma-separated decimal S0–S26 values.
`serial_x_code`	u8 0–4	`4`	Saved `ATX` level.
`serial_dtr_mode`	u8 0–3	`0`	Saved `AT&D` mode.
`serial_flow_mode`	u8 0–4	`0`	Saved `AT&K` mode.
`serial_dcd_mode`	u8 0–1	`1`	Saved `AT&C` mode.
`serial_stored_0` … `serial_stored_3`	string	empty	Stored phone-number slots for `AT&Zn=s` / `ATDSn`.

Input sanitisation

String values have embedded newlines and carriage returns stripped before being written back to disk. That means you cannot accidentally corrupt the config by pasting a multi-line value into the GUI.

4. GUI Configuration Console

Unless you have explicitly set `enable_console = false`, a GUI configuration window opens automatically when the gateway starts. It is backed by eframe/egui and provides live configuration editing plus a console that streams the server log in real time.

> **Headless mode**
> Systemd deployments, Docker images, or any other headless setup should run with `enable_console = false`. Otherwise the daemon will block waiting for a display server that does not exist.

Closing the GUI window does not stop the server — it continues running in headless mode. To stop the gateway, send it `Ctrl+C`, `SIGTERM`, or `SIGHUP` (see chapter 15).

4.1 Configuration Frames

The window is organised into three rows of two framed cards each, plus buttons and a console panel at the bottom:

- **Server** — toggle telnet/SSH, set the ports, set the session cap, set the idle timeout. A *More…* button reveals the Telnet Gateway options popup.
- **Security** — Require Login toggle, plus separate username and password fields for telnet and SSH.
- **File Transfer (XMODEM)** — transfer directory, negotiation timeout, per-block timeout, max retries.
- **AI Chat, Browser, and Weather** — Groq API key (rendered as a password field), browser homepage URL, and weather ZIP.
- **Serial Modem** — Enabled checkbox, port dropdown with a refresh (↺) button, baud, data bits, parity, stop bits, flow control. A *More…* button reveals the Hayes AT state popup.
- **General** — Verbose XMODEM Logging, Show GUI on Startup.

At the top of the window the title shows the gateway version and the local IP address that telnet clients will use to reach it. The IP is re-detected on every paint cycle, so if your DHCP lease changes the value updates automatically.

4.2 "More…" Popups

Advanced settings that are rarely changed live in popup windows opened from their frame's *More…* button.

4.2.1 Server — More

Mirrors the primary Server controls (so you can edit ports and toggles without leaving the popup) and adds two persistent outbound-gateway decisions that used to be asked interactively at dial time:

- **Telnet Gateway → Mode** — dropdown with *Telnet* or *Raw TCP*. Telnet is the default and parses IAC in both directions; Raw TCP disables the entire IAC layer for destinations that don't speak telnet at all. Sets `telnet_gateway_raw` in `xmodem.conf`.
- **Negotiate TTYPE / NAWS with remote** — checkbox, enabled only when Mode is *Telnet*. Turn on when you dial real telnet servers that want cooperative option negotiation. Sets `telnet_gateway_negotiate`.
- **SSH Gateway → Auth** — dropdown with *Key* or *Password*. Password is the default because it works out of the box with most remote accounts. When Key is selected the popup also displays the gateway's public key in a read-only text box below, ready to paste into the remote's `~/.ssh/authorized_keys`. Sets `ssh_gateway_auth`.

Both decisions take effect on the next gateway connection — no server restart is needed. The session-side Gateway Configuration menu (*Configuration → G* from the main telnet menu) edits the same two values.

> **Popup styling**
>
> Popups render on a deep forest-green panel with lighter-green text-entry fields so they're visually distinct from the navy main window. This is a UI convention — it doesn't affect any setting.

4.2.2 Serial Modem — More

Mirrors the primary serial port, baud, framing, and flow-control controls, and adds the full Hayes state that `AT&W` persists:

- **Echo (E1) / Verbose (V1) / Quiet (Q1)** — saved `ATE`, `ATV`, `ATQ` toggles.
- **Result level (X)** — 0 to 4. See ch. 9 for what each level emits.
- **DTR (&D)** — 0 to 3.
- **Flow (&K)** — 0 to 4 (modem-layer; hardware-wire flow is set by the main *Flow* dropdown above).
- **DCD (&C)** — 0 or 1.
- **S-Registers** — a comma-separated list of S0…S26 values, editable as free text. The format matches `serial_s_regs` in the config file.
- **Stored Phone Numbers** — four text fields for slots `&Z0` through `&Z3`. Leave empty for unset. `ATDSn` dials these.

4.3 Saving and Restarting

Each popup has its own **Save** button, which writes the current configuration to `xmodem.conf` *without restarting the server*. This is useful for changes that take effect on the next session — updating stored phone numbers, toggling verbose logging, or adjusting XMODEM timeouts — because no listener needs to be re-bound.

The main window also has a large **Save and Restart Server** button. Changes that affect how the listeners are bound — turning SSH on or off, changing a listen port, enabling or disabling the serial modem, changing credentials — need a restart to take effect. The button saves the config, sets an internal restart flag, and triggers an orderly shutdown. The main loop detects the flag and re-spawns the server; the GUI reopens after the restart completes.

> **Out-of-band changes**
> If a telnet or SSH session toggles a configuration value (for example the Telnet Gateway's *T* mode toggle, or the in-session Configuration menu), the GUI notices and refreshes its fields — unless you are currently editing them, in which case your edits are not clobbered.

4.4 Live Console Output

The bottom pane of the window shows the server log: new sessions, auth failures, dialup lookups, XMODEM progress when verbose is on, SSH host-key trust decisions, and shutdown messages. It auto-scrolls to the tail and caps at 2000 lines.

5. Telnet Access

The telnet server is on by default and listens on TCP port **2323**. Change the port via the GUI, the in-session *Configuration* menu, or by editing `telnet_port` in `xmodem.conf` and restarting.

5.1 Connecting

```
telnet 192.168.1.160 2323
```

Any standard telnet client works: `telnet`, PuTTY, SyncTERM, CGTERM on the Commodore 64, NetSwitcher on CP/M, a hardware WiFi modem plus a real C64, etc. The first thing the server does is negotiate a few telnet options via IAC (`WILL ECHO`, `WILL SGA`, `DO SGA`, `DO TTYPE`, `DO NAWS`), then prompt for terminal detection.

5.2 Terminal Detection

After the initial option handshake you will see:

```
Press BACKSPACE to detect terminal:
```

The byte you send determines the terminal class, as described in chapter 1. If the client already replied with an `IAC SB TTYPE IS …` subnegotiation during the initial handshake, the terminal is determined from that and the prompt is skipped entirely.

You are then asked to enable colour:

```
Use ANSI color? (Y/N):
```

No default: press `Y` or `N`.

5.3 Session Limits

- **Concurrent sessions** — capped by `max_sessions` (default 50).
- **Idle timeout** — `idle_timeout_secs` seconds (default 900 = 15 minutes). Set to `0` to disable. Idle timeouts apply to every prompt the server offers, including login, so users on slow modems should raise this.
- **Screen layout** — 40 columns on PETSCII, 80 on ANSI/ASCII, 22 rows of content (the remaining 2 rows are the header and prompt). Enforced by unit tests to avoid layout regressions.

5.4 Network Gating (Unauthenticated Mode)

When `security_enabled` is `false` (the default), the telnet server refuses connections from public IP addresses. Only the RFC 1918 private ranges — `10.0.0.0/8`, `172.16.0.0/12`, `192.168.0.0/16`, plus loopback, link-local, IPv6 loopback (`::1`), IPv6 link-local (`fe80::/10`),

and IPv6 unique local (`fd00::/8`) — may connect without authentication. Gateway addresses ending in `.1` are also rejected so a misconfigured NAT cannot accidentally bridge a telnet session through your router.

To accept connections from any IP address, enable security and configure a strong username and password.

5.5 The Main Menu

```
A  AI Chat
B  Simple Browser
C  Configuration
F  File Transfer
R  Troubleshooting
S  SSH Gateway
T  Telnet Gateway
W  Weather
X  Exit
```

6. SSH Access

The SSH interface is the recommended way to connect to the gateway from outside your trusted LAN. It is *not* enabled by default — it runs on port **2222**, uses its own username and password separate from the telnet credentials, and exposes the same menu system as telnet.

6.1 Enabling the SSH Server

Toggle the *SSH* checkbox in the GUI Server frame, or edit `xmodem.conf`:

```
ssh_enabled  = true
ssh_port     = 2222
ssh_username = admin
ssh_password = pick-something-strong
```

Then click *Save and Restart Server*, or restart manually. On first start with SSH enabled, the gateway generates an Ed25519 host key and persists it to `xmodem_ssh_host_key` next to the binary (mode `0600` on Unix). Subsequent starts reuse the same key so clients can verify the server's identity.

6.2 Connecting

```
ssh admin@192.168.1.160 -p 2222
```

The first time your client connects it will display the server's SHA-256 fingerprint and ask whether to trust it. Accept, and the key is recorded in your client's `~/.ssh/known_hosts` as usual.

After authentication you land on the same XMODEM Gateway menu as a telnet user, rendered in ANSI terminal mode. All features — file transfer, SSH gateway, telnet gateway, browser, AI chat, modem emulator, weather — are available.

6.3 Public-Key Authentication

The built-in SSH server supports password authentication only. However, the *outbound* SSH gateway (chapter 10) offers Ed25519 public-key auth to remote hosts — a separate concept.

> **Plaintext credentials**
> SSH credentials in `xmodem.conf` are stored in plaintext. The SSH connection itself is encrypted, but the file is not. Protect it with the `0600` permissions the gateway sets automatically, and do not reuse passwords from other services.

7. Security

Security is optional. In the default configuration the gateway trusts anyone on your private LAN and refuses public-Internet connections. Turn security on when you want explicit authentication, or when you need to accept connections from public addresses over SSH.

7.1 Enabling Login

Check the *Require Login* box in the GUI Security frame, or set:

```
security_enabled = true
username = admin
password = your-password
ssh_username = admin
ssh_password = your-password
```

Telnet and SSH credentials are independent. Use different values if you want different access levels, or identical values for simplicity.

7.2 Credential Comparison

Passwords are compared byte-for-byte using a constant-time equality check (`constant_time_eq`) so an attacker cannot measure tiny timing differences to recover the password one character at a time. This matters less over telnet (because the password is transmitted in cleartext anyway) but it is a genuine defence for SSH logins.

7.3 Per-IP Lockout

The telnet and SSH servers share a single in-memory lockout table keyed on the remote IP address. After **3** failed authentication attempts within 5 minutes the IP is blocked across *both* protocols for 5 minutes. An attacker cannot bounce between protocols to reset the counter.

Once the block expires the counter resets on the next successful login. Lockout events are logged to the console panel so a legitimate user who locked themselves out can see what happened.

7.4 File Permissions

On Unix, every file the gateway writes that may contain secrets is set to mode `0600` (owner read/write, no access for group or other):

- `xmodem.conf` — telnet password, SSH password, Groq API key.
- `dialup.conf` — no secrets, but dial-up mappings.
- `gateway_hosts` — known host fingerprints for outbound SSH dials.
- `xmodem_ssh_host_key` — the SSH server's private host key.
- `xmodem_gateway_ssh_key` — the gateway's outbound SSH client key.

Config file writes are done to a per-PID temporary filename and then renamed into place. This closes a TOCTOU window against symlink-swap attacks in shared working directories.

7.5 Audit Logging

Important events are written to the live console panel (and to stderr in headless mode):

- Auth successes and failures, with the remote IP.
- Lockout events when an IP hits the 3-failure threshold.
- SSH Gateway host-key trust decisions — TOFU accept, key update, and rejection — with host, port, key algorithm, and SHA-256 fingerprint. This gives you a forensic trail if a changed host key turns out to be a man-in-the-middle attempt.

8. File Transfer

File transfer is the original reason this project exists. The gateway implements three related protocols:

XMODEM (classic, 128-byte blocks)
: The original 1977 protocol. Uses a `SOH` header, a block number and its complement, 128 bytes of payload, and either an 8-bit checksum or a CRC-16. Compatible with every retro terminal program ever made.

XMODEM-1K (1024-byte blocks)
: A 1980s extension that uses an `STX` header and 1024 bytes of payload. Reduces protocol overhead by a factor of eight. Senders may mix 1K and 128-byte blocks freely within a single transfer; the gateway branches on the header byte per block.

YMODEM (batch + metadata)
: XMODEM-1K plus a block-0 header. Block 0 has `SOH`, block number `0`, complement `0xFF`, and a 128-byte payload containing the filename, size, and (optionally) mtime. The receiver uses the size to truncate trailing `SUB` padding exactly.

8.1 The Transfer Directory

All transfers are scoped to the `transfer_dir` setting (default `transfer`), relative to the gateway's working directory. The gateway canonicalises the path on every access to prevent symlinks from escaping the jail, and filenames are validated to reject path traversal.

Filename rules:

- Letters, digits, dots, hyphens, and underscores only.
- Maximum 64 characters.
- Cannot start with a dot.
- Cannot contain `..`.
- Must include at least one letter or digit.

File size is capped at **8 MB**. Larger uploads are aborted server-side with three `CAN` bytes.

8.2 Navigation

From the main menu press `F`:

```
U  Upload           (receive file from your terminal)
D  Download         (send file to your terminal)
X  Delete           (remove a file from the transfer dir)
C  Change directory (subdirectory within transfer_dir)
I  Toggle IAC escaping (for binary files with 0xFF bytes)
```

8.3 Uploads — Auto-Detection

Press `U`, enter a filename, and the server prints:

```
Begin XMODEM send now.
```

Now start the send from your terminal program (XMODEM, XMODEM-1K, or YMODEM). The gateway auto-detects which protocol the sender is using. The logic is:

1. The receiver first sends `'C'` (0x43) for CRC-16 mode. It retries every three seconds for two thirds of the negotiation timeout. Most modern clients respond to this.
2. If no response comes, the receiver falls back to sending `NAK` (0x15) to request checksum mode. This rescues very old clients that never implemented CRC.
3. The first header byte from the sender determines the block size: `SOH` (0x01) means a 128-byte block, `STX` (0x02) means a 1024-byte XMODEM-1K block.
4. If the first header is `SOH` with block number `0` and complement `0xFF`, it is a **YMODEM block 0**. The gateway reads the filename header, parses the file size, validates the CRC, `ACK`s, and sends a second `'C'` to start the data phase.
5. Within a single transfer the sender may mix 128-byte and 1024-byte blocks freely. Each block's header byte selects the payload length.
6. Transfer ends on `EOT` (0x04), which the server `ACK`s. For YMODEM, the server then sends `'C'` and consumes the end-of-batch "null block 0" the sender emits to signal "no more files" (Forsberg §7.4).
7. Once the transfer is complete, the file is truncated: YMODEM uses the exact size field from block 0 so files that legitimately end in `0x1A` bytes (EXEs, some archives) aren't corrupted; plain XMODEM and XMODEM-1K strip trailing `SUB` (0x1A) padding since there's no length metadata.

Cancellations are handled both ways: sending `ESC` from your side aborts the transfer, and three `CAN` bytes from either side abort with "Transfer cancelled".

> **Duplicate filename**
> If the name you type is already in the transfer directory, the server asks `Overwrite? (Y/N)` *before* starting the transfer. Answering `N` cancels cleanly; answering `Y` proceeds and truncates the existing file on save. This prevents a whole transfer from running only to fail at the final write step.

8.4 Downloads — Protocol Selection Prompt

Press `D`, then pick a file from the numbered list (paginated, 10 per page). Before the transfer starts, the gateway asks which protocol to use:

```
────────────────────────────────────
  SELECT PROTOCOL
────────────────────────────────────

  X  XMODEM       (128-byte blocks)
  1  XMODEM-1K    (1024-byte blocks)
  Y  YMODEM       (filename + size header, 1K)

  Pick one, or ESC to cancel:
```

Pick the protocol that your terminal program supports. The choice is immediate — one key, no Enter required. ESC (or ← on PETSCII, byte 0x5F) cancels.

The selection affects what the sender does:

- **XMODEM** — the sender transmits 128-byte `SOH` blocks exclusively.
- **XMODEM-1K** — the sender transmits 1024-byte `STX` blocks, with an opportunistic 128-byte `SOH` fallback on the final short block.
- **YMODEM** — the sender begins with a block-0 `SOH 0x00 0xFF` header containing the filename and size, waits for `'C'` acknowledgement, then transmits 1K data blocks just like XMODEM-1K.

8.5 ZMODEM Autostart Detection

ZMODEM (Chuck Forsberg's 1986 protocol, with its famous automatic "rz" trigger) is **not** supported. However, many terminal programs start a ZMODEM send by emitting the autostart sequence `**\x18B`, `**\x18A`, or `**\x18C`. If the gateway's idle input loop sees that pattern, it:

1. Logs "ZMODEM autostart detected; sending abort".
2. Transmits the standard abort: five `CAN` bytes (0x18 × 5) followed by five `BS` bytes (0x08 × 5) to wipe any "rz\r" prompt.
3. Pauses 500 ms and drains any in-flight ZMODEM frames.
4. Displays a friendly notice: "ZMODEM autostart detected — aborting. ZMODEM is not yet supported by this server. Please use XMODEM, XMODEM-1K, or YMODEM from the File Transfer menu instead."

8.6 Telnet NVT Transforms

Telnet's Network Virtual Terminal (RFC 854) imposes two wire-level transforms on any byte stream, including file transfers over a telnet connection:

- **IAC escaping.** Byte `0xFF` is the IAC (Interpret As Command) marker. Any `0xFF` in data — including a CRC byte that happens to be `0xFF` — must be doubled on the wire as `0xFF 0xFF`. The receiver collapses the pair back to a single `0xFF`.
- **CR-NUL stuffing.** A bare `0x0D` (CR) in data must be sent on the wire as `CR NUL` (`0x0D 0x00`). The receiver strips the NUL. Without this, the telnet parser on the peer side treats the CR as a line-ending sentinel and consumes the following byte.

The gateway applies both transforms on send *and* receive for inbound telnet sessions automatically. SSH and serial sessions skip them — SSH has no in-band command byte, and serial streams aren't subject to NVT rules.

The `I` toggle in the File Transfer menu flips both transforms together for the current session. The default is *on* for telnet (NVT-compliant clients — PuTTY / ExtraPuTTY, Tera Term, C-Kermit, SecureCRT — all escape). Toggle it *off* only if your terminal program is a rare one that sends raw bytes during XMODEM without escaping.

> **What you'll see if NVT is wrong**
> An upload that runs fine for one or two dozen blocks and then CRC-fails repeatedly is usually the first block containing a raw `0xFF` or `0x0D` byte that isn't being escaped by one side. A download that stalls after 3–4 blocks with the client repeatedly sending `'C'` instead of `ACK` is the same thing on the send side.

8.7 Protocol Timeout Tunables

Three config keys control transfer behaviour:

Key	Default	Meaning
`xmodem_negotiation_timeout`	45 s	Maximum time the receiver will wait for the first `SOH`/`STX` header from the sender, and vice versa.
`xmodem_block_timeout`	20 s	Per-block timeout. If a block header does not arrive within this long after an `ACK`/`NAK`, the receiver retries.
`xmodem_max_retries`	10	Maximum `NAK` retries per block before the receiver cancels the transfer with three `CAN`s.

> **Verbose diagnostics**
> Set `verbose = true` to log every header byte, CRC result, block number, and timeout event. This is invaluable when a transfer fails on a specific block and you need to know which side hiccuped.

9. Serial / Hayes AT Modem Emulation

The modem emulator turns the gateway into a virtual Hayes Smartmodem on a physical serial port. Retro hardware — a Commodore 64 with an RS-232 cartridge, a CP/M machine, an AltairDuino 2SIO2, an RC2014 — connects as if the gateway were a real modem. `ATDT`, `ATH`, `+++`, all of it works.

The emulator speaks a full Hayes command set: `AT`, `ATZ`, `AT&F`, `AT&W`, `AT&V`, `ATE`, `ATV`, `ATQ`, `ATI` variants, `ATH`, `ATA`, `ATO`, `ATX`, `AT&C`, `AT&D`, `AT&K`, `AT&Z`, `ATD` (with `T`, `P`, `L`, `S` variants), `ATSn`, S-registers S0–S26, the `A/` repeat-last-command shortcut, and the `+++` escape sequence with S2/S12 guard-time semantics.

9.1 Enabling the Emulator

From the GUI Serial Modem frame:

1. Check *Enabled*.
2. Pick a port from the dropdown. If your port does not appear, plug in the adapter and click the refresh (↺) button.
3. Set the baud rate, data bits, parity, stop bits, and wire-level flow control.
4. Click *Save and Restart Server* so the serial thread restarts with the new settings.

You can also configure everything live from an in-session telnet / SSH user: *Configuration → Modem Emulator*. That menu has `E` to enable, `S` to detect and select a port, and submenus for baud / framing / flow control. Changes applied through the in-session menu take effect immediately with no restart.

> **Serial lockout protection**
> When you change parameters from a serial session (i.e. through the modem itself), the server asks for confirmation and *reverts* if you don't respond within 60 seconds. If your new baud rate is wrong, your serial terminal will stop echoing and you will simply wait out the timer — your old settings come back unchanged.

Typical C64-friendly parameters: 2400–9600 baud, 8N1, no flow control. CP/M systems usually want 9600 8N1. The AltairDuino 2SIO2 is commonly configured for 9600 8N1 as well; remember to press *stop* and *aux1 up* on the front panel to configure its serial ports.

9.2 Dialing

The emulator routes dial strings over TCP. There is no real phone line — the "call" is a `tokio::net::TcpStream::connect()`.

Command	Action
`ATDT host:port`	Connect to a remote telnet server. `ATDT` and `ATDP` are equivalent (there is no audible difference over TCP).
`ATDT xmodem-gateway`	Built-in shortcut — dials this gateway's own main menu, so the serial device can use all gateway features (file transfer, browser, AI chat, etc.) without a network detour.
`ATDP host:port`	Same as `ATDT` — pulse selector, ignored.
`ATDSn` (n = 0–3)	Dial stored number in slot *n*. Bare `ATDS` dials slot 0.
`ATDL`	Redial the last number.
`AT&Zn=s`	Store phone number or `host:port` in slot *n* (0–3). Preserves hostname case.

Dial-string modifiers (only honoured when the dial string looks like a phone number, i.e. digits and formatting characters):

Modifier	Effect
`,`	Pause for S8 seconds (default 2) before continuing. Each comma adds S8 seconds.
`W`	Wait for dial tone. Adds S6 seconds to the pre-delay.
`;`	After "connect", return to command mode instead of entering online data mode. Applies to hostnames too.
`*`, `#`	DTMF digits. Preserved in the dial target for phone-number lookup.
`P`, `T`, `@`, `!`	Pulse, tone, quiet-answer, hookflash selectors. Accepted, ignored.

Modifiers are *not* stripped from hostnames. Dialling `ATDT pine.example.com` connects to "pine.example.com" verbatim — the heuristic notices that the dial string contains letters and dots and skips modifier processing for everything except the trailing `;`.

9.2.1 Examples

```
ATDT xmodem-gateway
CONNECT 9600
(local gateway main menu appears)

ATDT 192.168.1.160:2323
CONNECT 9600
(remote gateway appears)

ATDT bbs.example.com:23
CONNECT 9600

AT&Z0=bbs.example.com:23
OK
ATDS0
CONNECT 9600

AT&Z1=Pine.Example.com:22
OK
(note: hostname case is preserved)

ATDT 5551234
(looked up in dialup.conf, or NO CARRIER if unmapped)
```

9.2.2 Dial-up Mapping

The gateway also maintains `dialup.conf`, which maps phone numbers to `host:port` targets. Edit it through the Modem Emulator menu's *D* submenu. A built-in entry, `1001000`, always maps to the local gateway menu (equivalent to `ATDT xmodem-gateway`) and cannot be deleted. Numbers are matched by digits only — dashes, spaces, and parentheses are ignored — so `555-1234` and `5551234` hit the same entry.

9.3 Identification (ATI)

Command	Response
`ATI` / `ATI0`	Product ID and version: `XMODEM Gateway Modem Emulator v0.3.3`
`ATI1`	ROM checksum: `000` (virtual)
`ATI2`	ROM self-test: `OK`
`ATI3`	Firmware identifier: `XMODEM Gateway 0.3.3`
`ATI4`	Product description: `Hayes-compatible virtual modem over TCP`
`ATI5`	OEM / country code: `B00`
`ATI6`	Link diagnostics: `No link diagnostics available`
`ATI7`	Product info: `Product: xmodem-gateway (software emulator)`

9.4 Reset & Profile Storage

Command	Action
`ATZ`	Reset to the stored profile (read from `xmodem.conf`). Drops any active connection.
`AT&F`	Factory reset to gateway-friendly defaults. Does *not* overwrite the stored profile.
`AT&W`	Save current settings to `xmodem.conf`. Persists echo, verbose, quiet, all 27 S-registers, `X`, `&C`, `&D`, `&K`, and the four stored-number slots.
`AT&V`	Display current configuration — one line of mode flags, a second line listing every S-register, and a third line listing stored numbers.

9.5 Echo, Verbose, Quiet

Command	Effect
`ATE0` / `ATE1`	Echo off / on (default on).
`ATV0` / `ATV1`	Numeric / verbose result codes (default verbose).
`ATQ0` / `ATQ1`	Result codes on / silent mode (default on).

In verbose mode results are text (`OK`, `CONNECT`, `NO CARRIER`, `ERROR`, `RING`, `BUSY`, `NO DIALTONE`, `NO ANSWER`). In numeric mode they are digits. Quiet mode suppresses results altogether.

9.5.1 ATX Levels

Level	Extended codes	CONNECT format
`X0`	Basic only; BUSY / NO DIALTONE / NO ANSWER collapse to NO CARRIER	`CONNECT` (code 1)
`X1`	Basic + baud in CONNECT	`CONNECT <baud>` (code per baud)
`X2`	Adds NO DIALTONE detection	`CONNECT <baud>`
`X3`	Adds BUSY detection	`CONNECT <baud>`
`X4` (default)	Full extended set	`CONNECT <baud>`

Numeric CONNECT codes follow Hayes conventions: 1 = 300 baud, 5 = 1200, 10 = 2400, 12 = 9600, 16 = 19200, 28 = 38400, 87 = 115200. Non-standard baud rates report code 1.

9.6 DTR, Flow Control, and DCD

Command	Meaning
`AT&C0`	DCD always asserted.
`AT&C1` (Hayes default)	DCD reflects carrier state.
`AT&D0` (*gateway default*)	Ignore DTR entirely.
`AT&D1`	Drop DTR → return to command mode.
`AT&D2` (Hayes default)	Drop DTR → hang up.
`AT&D3`	Drop DTR → hang up and reset.
`AT&K0` (*gateway default*)	No modem-layer flow control.
`AT&K1`	Reserved.
`AT&K3`	Bidirectional RTS/CTS hardware flow control.
`AT&K4`	Bidirectional XON/XOFF software flow control.

Gateway-friendly default deviations

The emulator deviates from three Hayes defaults because retro clients rarely drive the relevant signals correctly:

- `AT&D0` instead of `&D2` — many C64, CP/M, and similar clients don't wire DTR. `&D2` would cause spurious disconnects.
- `AT&K0` instead of `&K3` — retro clients rarely implement hardware flow control. Physical-wire flow is still controlled by the `serial_flowcontrol` setting.
- `S7=15` instead of `S7=50` — keeps failed TCP dials responsive. Internally capped at 60 seconds.

All three can be overridden interactively (e.g. `AT&D2`, `AT&K3`, `ATS7=50`) and persisted with `AT&W`.

Signal pins are not driven

`AT&C`, `AT&D`, and `AT&K` are parsed, stored, persisted, and shown by `AT&V` — but the emulator does not drive the corresponding RS-232 hardware pins. DCD, RI, DSR, and DTR are not controlled, and RTS/CTS handshaking on the wire is governed by `serial_flowcontrol` rather than `&K`. Most retro terminal software works fine without these signals; if your program requires DCD before communicating, look for a "force DTR" or "ignore carrier" option.

9.7 S-Registers

Query with `ATSn?`, set with `ATSn=v`. Values are 0–255. `ATS?` prints a help table. `AT&W` saves all 27 values to `xmodem.conf`; `ATZ` restores them from the stored profile; `AT&F` resets to factory defaults.

Register	Default	Description
S0	5	Auto-answer ring count (0 disables).
S1	0	Current ring counter.
S2	43	Escape character (+). Values > 127 disable escape detection.
S3	13	Carriage return character.
S4	10	Line feed character.
S5	8	Backspace character.
S6	2	Wait-for-dial-tone seconds (used by `W` modifier).
S7	**15**	Wait-for-carrier seconds. Hayes default is 50; gateway reduces it for responsive TCP dials. Capped internally at 60.
S8	2	Comma pause time (seconds).
S9	6	Carrier detect response time (1/10 s).
S10	14	Carrier loss disconnect time (1/10 s).
S11	95	DTMF tone duration (ms).
S12	50	Escape-sequence guard time (1/50 s). 50 = 1.0 s. Set to 0 to disable escape detection.
S13 – S24	0	Reserved placeholders. Stored and persisted so legacy init strings don't halt with `ERROR`, but they have no effect.
S25	5	DTR detect time (1/100 s). Reserved — no DTR pin.
S26	1	RTS-to-CTS delay (1/100 s). Reserved — no RTS/CTS pins.

> **Do not collide S3/S4/S5**
> Command-mode line editing dispatches on the raw byte and checks the CR branch before the BS branch. Setting `S3 = 8` would make backspace terminate the line. Keep S3/S4/S5 distinct (the Hayes defaults 13/10/8 do the right thing).

9.8 The +++ Escape Sequence

From online (data-transfer) mode, type `+++` surrounded by at least **S12 × 1/50** seconds of silence on either side to return to command mode without hanging up. Once in command mode, `ATO` resumes the connection and `ATH` hangs up. This follows the Hayes guard-time spec exactly.

The implementation details:

- Escape character is `S2` (default 43 = `+`).
- Guard time is `S12 / 50` seconds on both sides of the triple.
- Setting `S12 = 0` or `S2 > 127` disables escape detection entirely.
- Fewer or more than three consecutive escape characters are passed through as data.
- The escape is interrupted if the sender emits any non-escape byte mid-sequence.

9.9 A/ — Repeat Last Command

`A/` (capital A, forward slash) with no `AT` prefix and no carriage return repeats the last command you entered. Real Hayes modems recognise this as a special two-character command. Useful for re-dialling or re-sending the last `ATZ` while debugging.

9.10 Ring Emulator

Telnet and SSH users can simulate an incoming phone call to the serial device from the Modem Emulator menu (`I` — Ring emulator). The modem sends `RING` to the serial port at standard US cadence (every 6 seconds). After `S0` rings (default 5) the modem auto-answers and the serial device receives the XMODEM Gateway main menu, just as if it had dialled in with `ATDT xmodem-gateway`. The serial device can also answer manually with `ATA` during ringing.

9.11 Unknown AT Commands

Commands the emulator has no hardware to implement — `ATB`, `ATC`, `ATL` (speaker loudness), `ATM` (speaker mute), `AT&B`, `AT&G`, `AT&J`, `AT&S`, `AT&T`, `AT&Y`, etc. — are accepted and return `OK`. This keeps legacy init strings working instead of halting mid-setup with `ERROR`.

10. SSH Gateway (Outbound)

The SSH Gateway proxies your inbound telnet or SSH session through to a remote SSH server. Useful for accessing SSH boxes from a Commodore 64 or any other client that can only speak telnet.

10.1 Using It

From the main menu, press `S`. The gateway shows the currently-configured auth mode, then prompts for the remote host, port, and username:

```
Auth: password
Host: example.com
Port (22): 22
Username: alice
```

Auth mode is set by `ssh_gateway_auth` in the config (default `password`). You can change it either from the GUI's *Server → More* popup or from the telnet session's *Configuration → Gateway Configuration* menu. The two modes are mutually exclusive — the gateway never silently falls back from one to the other, so auth failures are unambiguous:

- **Password** — the gateway asks for the remote account's password after the username prompt. Works with any SSH server that allows password auth.
- **Key** — the gateway offers its own Ed25519 keypair (`xmodem_gateway_ssh_key`, generated on first use and persisted at mode `0600`). The remote must have the matching public half in `~/.ssh/authorized_keys` first. The public key is shown in the GUI's Server → More popup when Auth is set to Key; you can also extract it on the command line with `ssh-keygen -y -f xmodem_gateway_ssh_key`.

Press `ESC` twice to disconnect.

10.2 Host-Key Trust (TOFU)

The first time you dial a given host, the gateway shows the server's SHA-256 fingerprint and asks whether to trust it. Accepted fingerprints are persisted to `gateway_hosts` (mode `0600`) and checked on every subsequent dial. A changed key triggers a prominent warning:

```
WARNING: HOST KEY HAS CHANGED!
Previous fingerprint: SHA256:...
Current  fingerprint: SHA256:...
Update (U), Abort (any other key):
```

All trust decisions — first accept, update, and reject — are written to the server log (and the GUI console) so you have a forensic trail if a changed key turns out to be a man-in-the-middle attempt.

10.3 Output Filtering

For PETSCII and ASCII terminals, ANSI escape sequences from the remote are automatically stripped and text is converted to the appropriate encoding. ANSI terminals get the raw output unmodified. PTY size is set to 40×25 for PETSCII and 80×24 for ANSI/ASCII.

10.4 Key File Summary

File	Role	Analogous to
`gateway_hosts`	Remote host fingerprints accepted by the SSH Gateway.	`~/.ssh/known_hosts`
`xmodem_ssh_host_key`	Host key the SSH *server* presents to incoming clients.	`/etc/ssh/ssh_host_ed25519_key`
`xmodem_gateway_ssh_key`	Outbound SSH client keypair used for pubkey auth to remote hosts.	`~/.ssh/id_ed25519`

11. Telnet Gateway (Outbound)

Press `T` from the main menu to dial a remote telnet server or any raw-TCP service on port 23 (BBSes, MUDs, legacy hand-rolled software).

11.1 Protocol Mode

The gateway displays the currently-configured mode on entry and proceeds straight to the host prompt. The mode is a persistent setting, not a per-session choice:

```
Mode: Telnet protocol
Host:
```

Change the mode from the GUI (*Server → More* popup, Telnet Gateway Mode dropdown) or from the telnet session's *Configuration → Gateway Configuration* menu. Changes take effect on the next dial — no restart needed.

11.2 The Three Modes Explained

11.2.1 Reactive mode (default)

`telnet_gateway_negotiate = false` and `telnet_gateway_raw = false`. The gateway parses IAC in both directions but does not send proactive option offers. It silently accepts cooperative `WILL ECHO` from the remote (so BBSes that expect the server to echo work correctly), refuses every other peer-initiated option with `DONT` / `WONT`, escapes outbound `0xFF` data bytes as `IAC IAC`, and parses inbound IAC so protocol bytes don't leak to the user's terminal. Works with any real telnet server *and* with raw-TCP services on port 23 that never initiate telnet options.

11.2.2 Cooperative mode

`telnet_gateway_negotiate = true`. In addition to reactive mode's behaviour, the gateway sends proactive `IAC WILL TTYPE`, `IAC WILL NAWS`, and `IAC DO ECHO` at connect time. On `SB TTYPE SEND` it replies with the user's terminal type (`PETSCII`, `ANSI`, or `DUMB`). On `DO NAWS` it replies with the user's actual window dimensions, and it forwards in-session NAWS updates if the user resizes their terminal. Option state is tracked through a full RFC 1143 six-state Q-method, so a mind-change while a prior offer is in flight resolves cleanly.

Enable this when dialling modern BBSes — you get echo cooperation, correct terminal-type adaptation, and full-screen UI at your actual window size.

11.2.3 Raw TCP

`telnet_gateway_raw = true`. The gateway disables its entire telnet-IAC layer: no IAC escaping on outbound, no IAC parsing on inbound. Supersedes `telnet_gateway_negotiate`. Intended for destinations that aren't really telnet — some legacy MUDs, hand-rolled BBS software, or services that happen to run on port 23 but speak some other byte-level protocol.

Bytes written *to* the local user are still IAC-escaped so their telnet client does not misinterpret a stray `0xFF` as a protocol byte.

11.3 Dialing

After the mode prompt, enter the host and port:

```
Host: bbs.example.com
Port (23): 23
```

Press `ESC` twice to disconnect. For PETSCII and ASCII terminals, ANSI escape sequences from the remote are filtered out. The gateway also has an 8 KiB cap on subnegotiation body length, so a malicious remote cannot exhaust memory by sending huge `SB` bodies without a terminating `IAC SE`.

12. AI Chat

The AI Chat feature sends prompts to the Groq API and displays the responses in the terminal. It is free to use — you need a Groq account and an API key — and the model is `llama-3.3-70b-versatile`.

12.1 Setup

1. Go to `https://console.groq.com/keys` and create a free account.
2. Generate an API key. It starts with `gsk_`.
3. In the GUI, paste the key into the *API Key* field (it displays as dots), or edit `xmodem.conf` and set `groq_api_key = gsk_your_key_here`.
4. Save. No restart needed.

12.2 Using AI Chat

From the main menu, press `A`. Type a question and press Enter. The server shows *Thinking…* while waiting for the response (timeout 30 seconds). The answer is paginated:

- `N` / `P` — next / previous page.
- `Q` — return to the main menu.
- From the answer screen, type a new question to continue the conversation.

Responses are word-wrapped to fit the terminal — 38 columns for PETSCII, 78 for ANSI/ASCII. Long code blocks and unusual punctuation are preserved but may wrap awkwardly on narrow screens.

> **No API key?**
> If `groq_api_key` is empty and the user presses `A`, the gateway displays instructions for obtaining a key instead of attempting the API call.

13. Web Browser

The text-mode browser renders HTML as plain text with numbered link markers. It works on all terminal types, including 40-column PETSCII.

13.1 Opening a Page

From the main menu, press `B`. The homepage loads automatically if `browser_homepage` is set (default `http://telnetbible.com`). You can also enter a URL or search query:

- A URL without a scheme — `example.com` — gets `https://` prepended automatically.
- Text without dots is treated as a search query and sent to DuckDuckGo.
- A `gopher://` URL uses the Gopher protocol.

13.2 Following Links

Links are marked with numbered tags: `[1]`, `[2]`, `[3]`. Press `L` and enter a number to follow.

13.3 Navigation Commands

Key	Action
`N` / `P`	Next / previous page.
`T` / `E`	Jump to top / end.
`L`	Follow link by number.
`G`	Go to a new URL or search.
`S`	Search for text in the page.
`F`	Fill out and submit forms.
`K`	Save as bookmark; on the home screen, opens the bookmarks list.
`B`	Back to the previous page.
`R`	Reload.
`H`	Help.
`Q`	Close page (back to browser home).
`ESC`	Exit to main menu.

13.4 Bookmarks and Forms

Up to 100 bookmarks are saved in `bookmarks.txt` next to the binary. When a page has forms, press `F` to interact — pick a form, edit fields by number, `S` to submit or `Q` to cancel.

13.5 Limits

- Maximum page size: 1 MB.
- Maximum rendered lines: 5000.
- HTTP request timeout: 15 s.
- Page history depth: 50.
- HTTPS connections that fail due to TLS errors retry over HTTP.

14. Weather

The Weather feature displays the current conditions and a 3-day forecast for any US ZIP code. Data is pulled from Open-Meteo (`https://open-meteo.com`), which is free and requires no API key.

1. From the main menu, press `W`.
2. Enter a 5-digit US ZIP code, or press Enter to accept the last-used one.
3. Current temperature, humidity, wind speed and compass direction, plus a 3-day forecast are displayed.

The last-used ZIP is persisted to `weather_zip` in `xmodem.conf` so subsequent calls default to the same location.

15. Signal Handling & Operations

The gateway installs POSIX signal handlers at startup using the `signal-hook` crate. On receipt of any of `SIGINT`, `SIGTERM`, or `SIGHUP`, the server begins an orderly shutdown.

Signal	Typical source	Effect
`SIGINT`	`Ctrl+C` from the terminal	Shut down. Connected sessions are notified with a broadcast message.
`SIGTERM`	`kill`, systemd	Shut down.
`SIGHUP`	systemd `reload`, terminal disconnect	Shut down. (The unit file maps *reload* to SIGHUP for familiar semantics.)

15.1 Shutdown Sequence

When a signal arrives:

1. An `AtomicBool` flag is set.
2. A watcher thread fires a tokio `Notify` to wake the async runtime.
3. Listeners stop accepting new connections.
4. A broadcast notification is sent to every active telnet and SSH session: "Server is shutting down. Goodbye!"
5. The serial modem thread exits cleanly, releasing the port.
6. The GUI window closes automatically.
7. The process exits.

15.2 Under systemd

```
sudo systemctl status xmodem-gateway   # current status
journalctl -u xmodem-gateway -f        # live log
sudo systemctl reload xmodem-gateway   # graceful stop (SIGHUP)
sudo systemctl restart xmodem-gateway  # restart
sudo systemctl stop xmodem-gateway     # stop (SIGTERM)
```

The unit's `Restart=on-failure` policy respawns the process if it exits with a non-zero code, rate-limited to 5 starts per 60 seconds via `StartLimitIntervalSec` / `StartLimitBurst`.

16. Troubleshooting

16.1 Telnet Session Refused (Public IP)

If an unauthenticated client sees "Connections from this address are not accepted", the source IP is outside the private ranges listed in chapter 5. Either connect from a private-network address or enable `security_enabled = true` and configure credentials.

16.2 Session Denied — Too Many Attempts

Per-IP lockout after three failures. Wait 5 minutes. The lockout covers both telnet and SSH — retrying on the other protocol will not reset the counter.

16.3 No Serial Port Detected

- Plug in the adapter before clicking the refresh (↺) button in the GUI.
- On Linux, make sure your user is in the `dialout` group (`sudo usermod -aG dialout "$USER"`), then log out and back in.
- On Linux under systemd, the unit file sets `PrivateDevices=yes` which hides most device nodes. If you need USB-serial access, remove that hardening line or switch to a raw `DeviceAllow=` rule for your exact adapter.
- On macOS, adapters typically show up as `/dev/tty.usbserial-XXXX` or `/dev/tty.usbmodemXXXX`.
- On Windows, check Device Manager. Ports appear as `COM3`, `COM4`, etc.

16.4 XMODEM Transfer Times Out

Turn on verbose mode (`verbose = true` in `xmodem.conf`, or the *Verbose XMODEM Logging* checkbox in the GUI General frame) and retry. The log will show every header byte, every CRC result, every timeout. Common causes:

- Client and server disagree on CRC vs. checksum — the client is very old and never implemented CRC. The gateway falls back to checksum automatically after two-thirds of the negotiation timeout.
- Client uses ZMODEM autostart. The gateway aborts it cleanly and shows a notice — pick XMODEM, XMODEM-1K, or YMODEM instead.
- IAC escaping mismatch when transferring binary files with `0xFF` bytes over telnet. Toggle it with `I` in the File Transfer menu.
- Slow or unreliable serial link — raise `xmodem_block_timeout` and `xmodem_max_retries`.

16.5 SSH Won't Connect

- Check `ssh_enabled = true` and that the server has been restarted since the change.
- Confirm the port with `ss -tlnp | grep 2222` on Linux.

- On first connect, accept the gateway's host-key fingerprint. Your client records it in `~/.ssh/known_hosts`.
- If your client complains about host-key mismatch, you have probably regenerated `xmodem_ssh_host_key`. Delete the old entry from `known_hosts` or compare fingerprints carefully.
- Public-key auth against the *server* is not supported — only password auth. Public-key auth is only available on the *outbound* SSH Gateway.

16.6 Modem Shows "NO CARRIER" on Every Dial

- Check the host and port — `ATDT host:port` must include the port unless the host resolves to something listening on 23.
- S7 defaults to 15 seconds. Networks that need longer dial times should `ATS7=50` (or up to 60).
- Verify the host is reachable from the gateway machine with `nc -zv host port` or `telnet host port`.
- If you dialled a phone number (all-digit string), check `dialup.conf` — unmapped numbers return `NO CARRIER`.

16.7 GUI Won't Open

- Check that `enable_console = true`.
- On headless servers, leave it at `false` — the GUI requires a display server (X11 or Wayland).
- On Linux, make sure `libudev`, `libxkbcommon`, and your display-server libraries are installed.
- Missing GPU support? The GUI uses `wgpu` via eframe. Software-only fallback via a different feature set would require a rebuild; most modern Linux distros work out of the box.

16.8 Stuck Serial Session After Parameter Change

If you change baud from a serial session with mismatched terminal settings, don't panic. The gateway waits up to 60 seconds for confirmation and reverts automatically on timeout. Within 60 seconds your old settings come back.

Appendix A — Configuration Key Reference

Every key recognised by `xmodem.conf`, sorted alphabetically. The parser is case-sensitive on keys but booleans (`true`/`false`) are compared case-insensitively. Unknown keys are silently ignored. Out-of-range numeric values fall back to the default.

Key	Type	Default	Range / validation	Description
`browser_homepage`	string	`http://telnetbible.com`	any URL or empty	Page loaded when the browser opens. E prompt.
`enable_console`	bool	`true`	true/false	Show GUI window at startup. Set false
`groq_api_key`	string	empty	any	Groq API key. Empty disables AI chat. S file.
`idle_timeout_secs`	u64	`900`	0 = disabled	Per-session idle timeout. Applies to ever
`max_sessions`	usize	`50`	≥ 1	Concurrent telnet-session cap.
`password`	string	`changeme`	non-empty	Telnet password. Stored plaintext.
`security_enabled`	bool	`false`	true/false	Require login on telnet and SSH. Also r telnet.
`serial_baud`	u32	`9600`	≥ 300	Serial baud rate.
`serial_databits`	u8	`8`	5, 6, 7, or 8	Data bits.
`serial_dcd_mode`	u8	`1`	0 or 1	Saved `AT&C` DCD mode.
`serial_dtr_mode`	u8	`0`	0–3	Saved `AT&D` DTR-handling mode.
`serial_echo`	bool	`true`	true/false	Saved `ATE` echo state.
`serial_enabled`	bool	`false`	true/false	Run the Hayes modem emulator on the
`serial_flow_mode`	u8	`0`	0–4	Saved `AT&K` modem-layer flow mode.
`serial_flowcontrol`	string	`none`	none / hardware / software	Wire-level flow control.
`serial_parity`	string	`none`	none / odd / even	Parity bit.
`serial_port`	string	empty	OS-specific path	Serial device. `/dev/ttyUSB0`, `COM3`, etc.
`serial_quiet`	bool	`false`	true/false	Saved `ATQ` quiet mode.
`serial_s_regs`	string	see note	CSV of 27 u8 values	S0–S26, comma-separated. Default: `5,0,43,13,10,8,2,15,2,6,14,95,50,0`
`serial_stopbits`	u8	`1`	1 or 2	Stop bits.
`serial_stored_0` … `serial_stored_3`	string	empty	any	Stored phone-number slots.
`serial_verbose`	bool	`true`	true/false	Saved `ATV` verbose/numeric mode.
`serial_x_code`	u8	`4`	0–4	Saved `ATX` result-code level.

Key	Type	Default	Range / validation	Description
`ssh_enabled`	bool	`false`	true/false	Run the SSH server.
`ssh_gateway_auth`	string	`password`	`key` or `password`	Auth mode for the outbound SSH Gatew
`ssh_password`	string	`changeme`	non-empty	SSH password. Independent of telnet. S
`ssh_port`	u16	`2222`	≥ 1	SSH listen port.
`ssh_username`	string	`admin`	non-empty	SSH username.
`telnet_enabled`	bool	`true`	true/false	Run the telnet server.
`telnet_gateway_negotiate`	bool	`false`	true/false	Outbound telnet gateway: send proactiv offers.
`telnet_gateway_raw`	bool	`false`	true/false	Outbound telnet gateway: raw-TCP esca
`telnet_port`	u16	`2323`	≥ 1	Telnet listen port.
`transfer_dir`	string	`transfer`	non-empty	Base directory for file transfers.
`username`	string	`admin`	non-empty	Telnet username.
`verbose`	bool	`false`	true/false	Detailed XMODEM protocol logging.
`weather_zip`	string	empty	5 digits	Last-used US ZIP. Updated automaticall
`xmodem_block_timeout`	u64	`20`	≥ 1 (seconds)	Per-block timeout during transfer.
`xmodem_max_retries`	usize	`10`	≥ 1	NAK retry cap per block.
`xmodem_negotiation_timeout`	u64	`45`	≥ 1 (seconds)	Initial negotiation timeout for XMODEM

Appendix B — AT Command Reference

Quick reference for every AT command the emulator understands. See chapter 9 for detailed explanations. All commands are case-insensitive. Multiple commands may be combined on a single `AT` line (e.g. `ATE0Q1V0`).

B.1 Core Commands

Command	Effect
`AT`	Attention. Returns `OK`.
`AT?`	Display AT command help.
`ATZ`	Reset to stored profile (saved by `AT&W`).
`AT&F`	Factory reset to gateway-friendly defaults.
`AT&W` / `AT&W0`	Save current settings to `xmodem.conf`.
`AT&V`	Display current configuration.
`ATH` / `ATH0`	Hang up any active connection.
`ATA`	Answer incoming ring.
`ATO` / `ATO0`	Return to online mode (after +++ escape).
`A/`	Repeat last command (no AT prefix, no CR).
`+++`	Escape to command mode (guard-time gated).

B.2 Identification

Command	Response
`ATI` / `ATI0`	Product ID and version.
`ATI1`	ROM checksum (`000`).
`ATI2`	ROM self-test (`OK`).
`ATI3`	Firmware identifier.
`ATI4`	Product description.
`ATI5`	OEM / country code (`B00`).
`ATI6`	Link diagnostics.
`ATI7`	Product info.

B.3 Echo, Verbosity, Quiet, Result Codes

Command	Effect
`ATE0`	Echo off.
`ATE1` (default)	Echo on.
`ATV0`	Numeric result codes.
`ATV1` (default)	Verbose (text) result codes.
`ATQ0` (default)	Result codes enabled.
`ATQ1`	Quiet mode — all results suppressed.
`ATX0`	Basic result codes only.
`ATX1`	Basic plus baud rate in CONNECT.
`ATX2`	Adds NO DIALTONE detection.
`ATX3`	Adds BUSY detection.
`ATX4` (default)	Full extended result codes.

B.4 Hardware Pin State

Command	Effect
`AT&C0`	DCD always asserted.
`AT&C1` (Hayes default)	DCD reflects carrier state.
`AT&D0` (*gateway default*)	Ignore DTR.
`AT&D1`	DTR drop → command mode.
`AT&D2` (Hayes default)	DTR drop → hang up.
`AT&D3`	DTR drop → hang up + reset.
`AT&K0` (*gateway default*)	No modem-layer flow control.
`AT&K3`	Bidirectional RTS/CTS.
`AT&K4`	Bidirectional XON/XOFF.

B.5 Dialing

Command	Effect
`ATD` *`target`*	Dial *target*. Bare `D` (no T/P).
`ATDT` *`host:port`*	Tone-dial. Default for all TCP dials.
`ATDP` *`host:port`*	Pulse-dial (identical to `ATDT` over TCP).
`ATDT xmodem-gateway`	Built-in shortcut: connect to this gateway's main menu.
`ATDL`	Redial last number.
`ATDS` / `ATDS0`	Dial stored number in slot 0.
`ATDS1` … `ATDS3`	Dial stored number in slots 1–3.
`AT&Zn=`*`s`*	Store phone number or host in slot *n* (0–3). Preserves hostname case.

Dial-string modifiers (phone-number context only, except `;` which applies to hostnames too):

Modifier	Effect
`,`	Pause for S8 seconds per comma.
`W`	Wait for dial tone (adds S6 seconds).
`;`	Return to command mode after connect.
`*`, `#`	DTMF digits, preserved for dial-up lookup.
`P`, `T`, `@`, `!`	Ignored.

B.6 S-Registers

Command	Effect
`ATS?`	Show S-register help.
`ATS`*`n`*`?`	Query S-register *n* (returns 3-digit decimal).
`ATS`*`n`*`=`*`v`*	Set S-register *n* to *v* (0–255).

Full register list is in chapter 9 (section 9.7).

B.7 Unknown Commands

Unrecognised AT commands return `OK` (silent-accept fallback). This covers commands the emulator has no hardware to implement: `ATB`, `ATC`, `ATL` (speaker loudness), `ATM` (speaker mute), `AT&B`, `AT&G`, `AT&J`, `AT&S`, `AT&T`, `AT&Y`, and others. The intent is to let legacy init strings run to completion rather than halt mid-setup with `ERROR`.

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